

GRouNdScale Model Scaling Report

Parameter Sizes, GPU Memory and Step Times across HVG Configurations
(`size-compute`)

Auto-generated from `size-compute` run logs

October 5, 2026

1 Overview

This report summarises results from 32 completed training runs of the Causal GAN pipeline with increasingly higher highly variable gene (HVG) counts, from 1,000 to 16,500 in steps of 500. Each run uses the same model architecture and training configuration (20 training steps for the causal controller, followed by 20 steps for the main causal GAN), differing only in the number of highly variable genes selected during preprocessing.

The 17000 HVG run(s) were incomplete or failed at the time this report was generated and are excluded from all tables and figures. In particular, the 17,000 HVG run encountered CUDA out-of-memory during generator initialization.

All included runs completed without CUDA out-of-memory errors. The largest included run (16,500 HVGs) reached a peak GPU memory of 79,719 MiB, approaching the GPU's 80 GB capacity.

2 GRN Characteristics

Table 1: Causal graph statistics for each HVG configuration. The TF/Target ratio is the number of transcription factors divided by the number of target genes.

HVGs	TFs	Target Genes	TF/Target Ratio (%)	Imposed Edges	Possible Edges	Edge Density (%)
1,000	63	928	6.79	13,920	58,464	23.8
1,500	89	1410	6.31	21,150	125,490	16.9
2,000	127	1873	6.78	28,095	237,871	11.8
2,500	161	2339	6.88	35,085	376,579	9.3
3,000	202	2798	7.22	41,970	565,196	7.4
3,500	237	3263	7.26	48,945	773,331	6.3
4,000	287	3713	7.73	55,695	1,065,631	5.2
4,500	324	4176	7.76	62,640	1,353,024	4.6
5,000	370	4630	7.99	69,450	1,713,100	4.1
5,500	414	5086	8.14	76,290	2,105,604	3.6
6,000	464	5536	8.38	83,040	2,568,704	3.2
6,500	506	5994	8.44	89,910	3,032,964	3.0
7,000	555	6445	8.61	96,675	3,576,975	2.7
7,500	599	6901	8.68	103,515	4,133,699	2.5
8,000	639	7361	8.68	110,415	4,703,679	2.3
8,500	685	7815	8.77	117,225	5,353,275	2.2
9,000	733	8267	8.87	124,005	6,059,711	2.0
9,500	774	8726	8.87	130,890	6,753,924	1.9
10,000	817	9183	8.90	137,745	7,502,511	1.8
10,500	851	9649	8.82	144,735	8,211,299	1.8
11,000	887	10113	8.77	151,695	8,970,231	1.7
11,500	918	10582	8.68	158,730	9,714,276	1.6
12,000	949	11051	8.59	165,765	10,487,399	1.6
12,500	974	11526	8.45	172,890	11,226,324	1.5
13,000	1002	11998	8.35	179,970	12,021,996	1.5
13,500	1027	12473	8.23	187,095	12,809,771	1.5
14,000	1045	12955	8.07	194,325	13,537,975	1.4
14,500	1064	13436	7.92	201,540	14,295,904	1.4
15,000	1098	13902	7.90	208,530	15,264,396	1.4
15,500	1118	14382	7.77	215,730	16,079,076	1.3
16,000	1131	14869	7.61	223,035	16,816,839	1.3
16,500	1146	15354	7.46	230,310	17,595,684	1.3

3 Training Step Times

Table 2: Average time per training step for the causal controller (first training phase) and causal generator (second training phase), extracted from the final step log line of each phase.

HVGs	Controller Avg Step (ms)	Generator Avg Step (ms)
1,000	37	87
1,500	42	134
2,000	36	161
2,500	38	206
3,000	37	240
3,500	41	263
4,000	37	296
4,500	42	302
5,000	39	363
5,500	39	391
6,000	28	399
6,500	40	471
7,000	40	502
7,500	29	524
8,000	40	558
8,500	40	593
9,000	43	638
9,500	33	670
10,000	42	693
10,500	42	737
11,000	35	762
11,500	41	805
12,000	39	840
12,500	41	864
13,000	39	909
13,500	42	940
14,000	42	979
14,500	44	1012
15,000	32	1051
15,500	35	1094
16,000	29	1127
16,500	40	1196

Table 3: Expected training duration for a full run of 200,000 causal-controller steps and 500,000 causal-generator steps, based on the average step times in Table 2.

HVGs	Controller (200,000 steps)	Generator (500,000 steps)	Total
1,000	2.1 h	12.1 h	14.1 h (0.6 d)
1,500	2.3 h	18.6 h	20.9 h (0.9 d)
2,000	2.0 h	22.4 h	24.4 h (1.0 d)
2,500	2.1 h	28.6 h	30.7 h (1.3 d)
3,000	2.1 h	33.3 h	35.4 h (1.5 d)
3,500	2.3 h	36.5 h	38.8 h (1.6 d)
4,000	2.1 h	41.1 h	43.2 h (1.8 d)
4,500	2.3 h	41.9 h	44.3 h (1.8 d)
5,000	2.2 h	50.4 h	52.6 h (2.2 d)
5,500	2.2 h	54.3 h	56.5 h (2.4 d)
6,000	1.6 h	55.4 h	57.0 h (2.4 d)
6,500	2.2 h	65.4 h	67.6 h (2.8 d)
7,000	2.2 h	69.7 h	71.9 h (3.0 d)
7,500	1.6 h	72.8 h	74.4 h (3.1 d)
8,000	2.2 h	77.5 h	79.7 h (3.3 d)
8,500	2.2 h	82.4 h	84.6 h (3.5 d)
9,000	2.4 h	88.6 h	91.0 h (3.8 d)
9,500	1.8 h	93.1 h	94.9 h (4.0 d)
10,000	2.3 h	96.2 h	98.6 h (4.1 d)
10,500	2.3 h	102.4 h	104.7 h (4.4 d)
11,000	1.9 h	105.8 h	107.8 h (4.5 d)
11,500	2.3 h	111.8 h	114.1 h (4.8 d)
12,000	2.2 h	116.7 h	118.8 h (5.0 d)
12,500	2.3 h	120.0 h	122.3 h (5.1 d)
13,000	2.2 h	126.2 h	128.4 h (5.4 d)
13,500	2.3 h	130.6 h	132.9 h (5.5 d)
14,000	2.3 h	136.0 h	138.3 h (5.8 d)
14,500	2.4 h	140.6 h	143.0 h (6.0 d)
15,000	1.8 h	146.0 h	147.8 h (6.2 d)
15,500	1.9 h	151.9 h	153.9 h (6.4 d)
16,000	1.6 h	156.5 h	158.1 h (6.6 d)
16,500	2.2 h	166.1 h	168.3 h (7.0 d)

4 Parameter Counts

Table 4: Parameter counts for each model component and its optimiser, from `params-size.json`, expressed in millions (M).

HVGs	Generator	Critic	Labeler	Anti-Lab.	Gen. Opt.	Crit. Opt.	Lab. Opt.	Anti-Lab. Opt.
1,000	4.12M	1.67M	10.01M	10.01M	2.59M	5.02M	30.00M	30.00M
1,500	5.89M	2.19M	11.03M	11.03M	3.93M	6.58M	33.05M	33.05M
2,000	7.60M	2.71M	12.03M	12.03M	5.22M	8.12M	36.05M	36.05M
2,500	9.32M	3.22M	13.03M	13.03M	6.52M	9.65M	39.05M	39.05M
3,000	11.03M	3.73M	14.03M	14.03M	7.80M	11.19M	42.05M	42.05M
3,500	12.75M	4.24M	15.03M	15.03M	9.09M	12.72M	45.05M	45.05M
4,000	14.42M	4.75M	16.03M	16.03M	10.35M	14.26M	48.05M	48.05M
4,500	16.14M	5.27M	17.03M	17.03M	11.64M	15.80M	51.06M	51.06M
5,000	17.83M	5.78M	18.03M	18.03M	12.90M	17.33M	54.06M	54.06M
5,500	19.52M	6.29M	19.03M	19.03M	14.17M	18.87M	57.06M	57.06M
6,000	21.20M	6.80M	20.03M	20.03M	15.43M	20.40M	60.06M	60.06M
6,500	22.90M	7.31M	21.03M	21.03M	16.71M	21.94M	63.06M	63.06M
7,000	24.58M	7.83M	22.03M	22.03M	17.96M	23.48M	66.06M	66.06M
7,500	26.28M	8.34M	23.03M	23.03M	19.23M	25.01M	69.06M	69.06M
8,000	27.98M	8.85M	24.03M	24.03M	20.52M	26.55M	72.06M	72.06M
8,500	29.67M	9.36M	25.03M	25.03M	21.78M	28.08M	75.06M	75.06M
9,000	31.36M	9.87M	26.03M	26.03M	23.04M	29.62M	78.06M	78.06M
9,500	33.06M	10.39M	27.03M	27.03M	24.32M	31.16M	81.06M	81.06M
10,000	34.76M	10.90M	28.03M	28.03M	25.59M	32.69M	84.06M	84.06M
10,500	36.48M	11.41M	29.03M	29.03M	26.89M	34.23M	87.06M	87.06M
11,000	38.20M	11.92M	30.03M	30.03M	28.18M	35.76M	90.06M	90.06M
11,500	39.92M	12.43M	31.03M	31.03M	29.49M	37.30M	93.06M	93.06M
12,000	41.65M	12.95M	32.03M	32.03M	30.80M	38.84M	96.06M	96.06M
12,500	43.40M	13.46M	33.03M	33.03M	32.12M	40.37M	99.06M	99.06M
13,000	45.13M	13.97M	34.03M	34.03M	33.44M	41.91M	102.06M	102.06M
13,500	46.88M	14.48M	35.03M	35.03M	34.76M	43.44M	105.06M	105.06M
14,000	48.64M	14.99M	36.03M	36.03M	36.11M	44.98M	108.06M	108.06M
14,500	50.40M	15.51M	37.03M	37.03M	37.45M	46.52M	111.06M	111.06M
15,000	52.12M	16.02M	38.03M	38.03M	38.74M	48.05M	114.06M	114.06M
15,500	53.88M	16.53M	39.03M	39.03M	40.08M	49.59M	117.06M	117.06M
16,000	55.65M	17.04M	40.03M	40.03M	41.44M	51.12M	120.06M	120.06M
16,500	57.42M	17.55M	41.03M	41.03M	42.79M	52.66M	123.06M	123.06M

5 GPU Memory Requirements

Table 5: Peak GPU memory during each training phase, determined by cross-referencing log timestamps with the `gpu-usage.csv` monitoring data. The controller plateau is the maximum memory during causal-controller training; the generator plateau is the maximum during the main GAN training phase.

HVGs	Controller (MiB)	Generator (MiB)
1,000	749	5,517
1,500	1,249	8,445
2,000	753	10,359
2,500	1,139	13,033
3,000	749	15,393
3,500	855	17,423
4,000	1,215	20,097
4,500	835	22,443
5,000	1,259	24,787
5,500	1,013	26,763
6,000	985	29,081
6,500	1,077	31,433
7,000	1,071	33,763
7,500	1,045	36,095
8,000	1,065	38,791
8,500	1,699	41,275
9,000	1,125	43,079
9,500	1,743	45,951
10,000	1,251	47,775
10,500	1,239	50,671
11,000	1,265	52,529
11,500	1,291	54,931
12,000	1,317	57,351
12,500	1,453	59,759
13,000	1,383	62,181
13,500	1,403	64,625
14,000	1,557	67,083
14,500	1,479	69,613
15,000	1,627	71,935
15,500	1,531	74,395
16,000	1,557	76,889
16,500	1,613	79,719

6 Scaling Graphs

6.1 Linear Fits for Causal-Generator Requirements

Using ordinary least-squares fits over the completed runs, with x denoting the number of highly variable genes, the observed causal-generator requirements are approximated by:

$$\widehat{M}_{\text{gen}}(x) = 4.7268x + 927.7 \text{ MiB}, \quad R^2 = 0.9998, \quad (1)$$

$$\widehat{t}_{\text{gen}}(x) = 0.06897x + 15.49 \text{ ms/step}, \quad R^2 = 0.9984. \quad (2)$$

The memory fit uses the peak generator-phase GPU memory in MiB; the time fit uses the average time from the second final-step log line. These fits describe the measured range and should not be interpreted as guarantees beyond it.

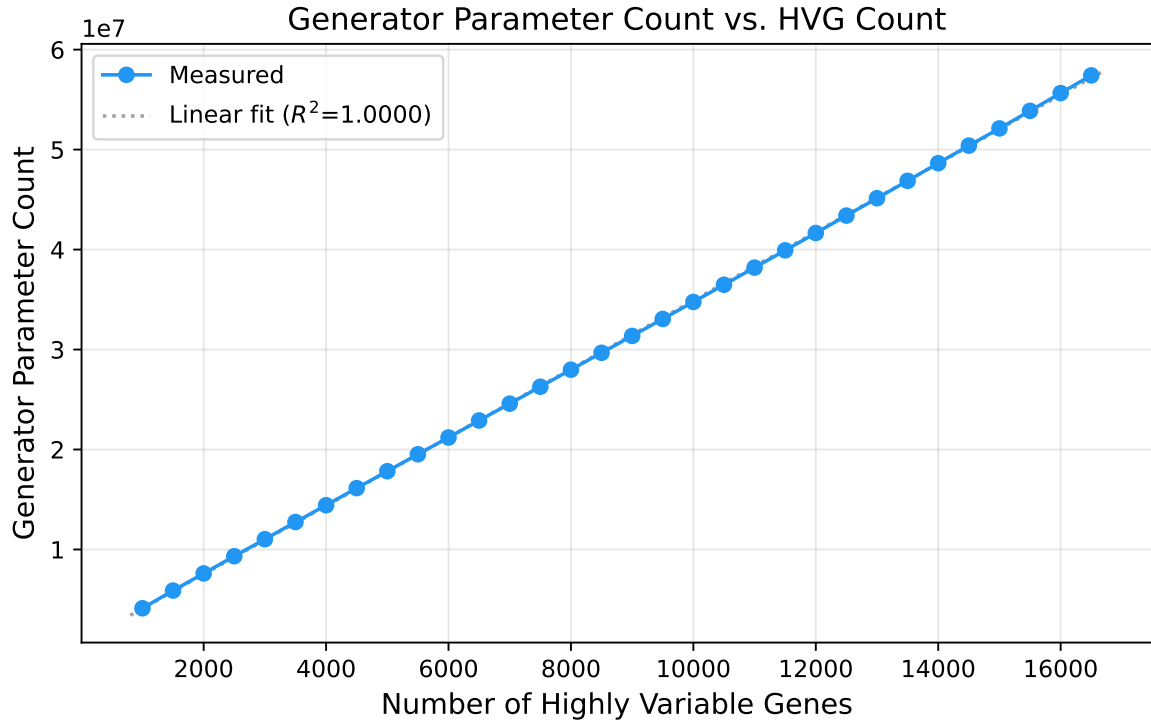


Figure 1: Generator parameter count as a function of the number of highly variable genes, fitted with $y = 3424.1 \cdot x + 670033$ ($R^2 = 1.0000$).

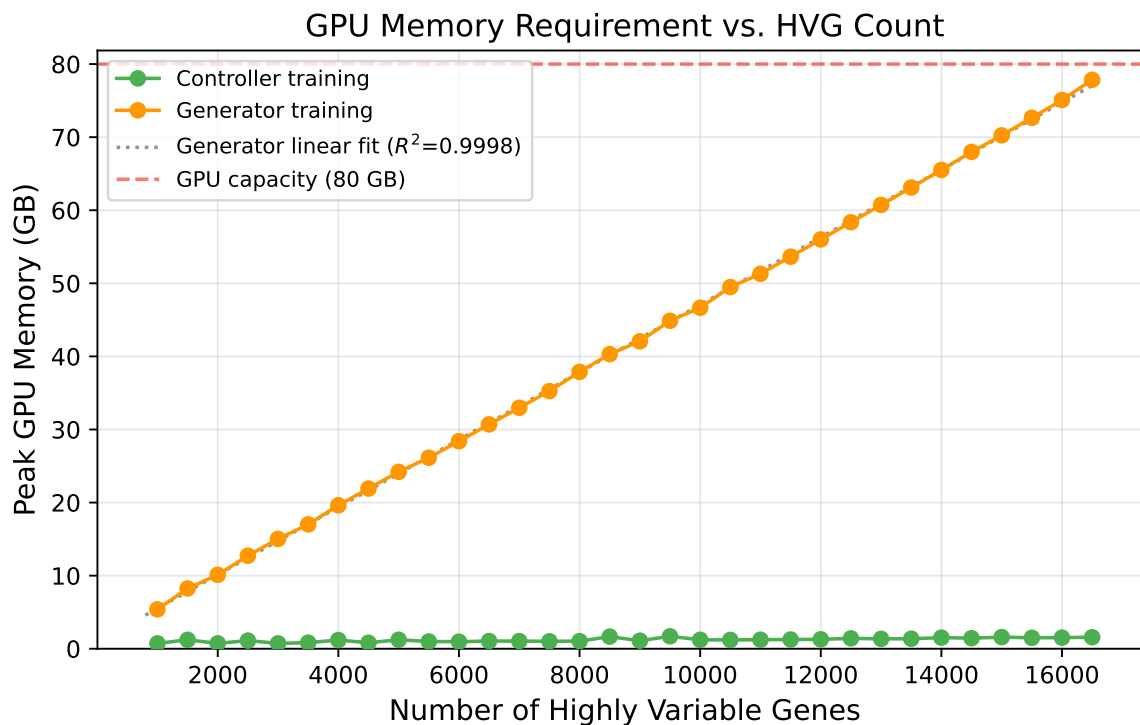


Figure 2: Peak GPU memory during controller and generator training as a function of HVG count. The red dashed line marks the GPU's total capacity (80 GB = 81 920 MiB). The largest completed run approaches the capacity limit.

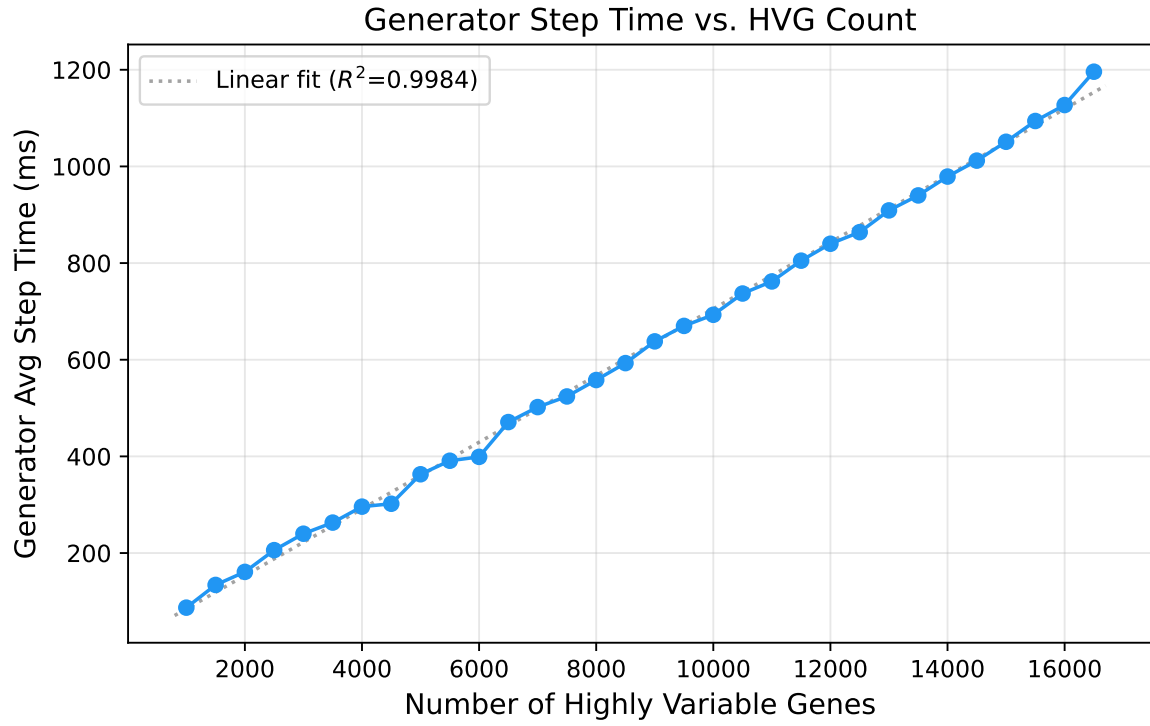


Figure 3: Causal generator average step time versus the number of highly variable genes. The dotted line is the linear fit used in the time equation above ($R^2 = 0.9984$).

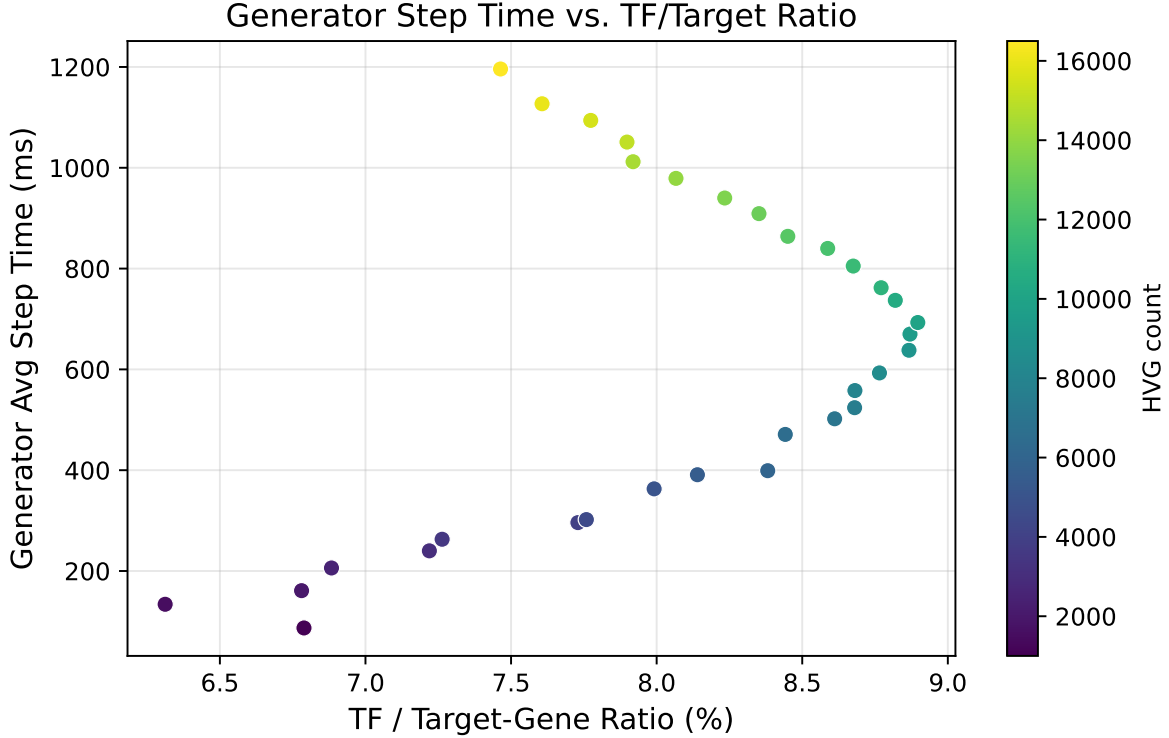


Figure 4: Causal generator average step time versus the TF/target-gene ratio. Point color indicates HVG count; the ratio is non-monotonic, peaking near 10,000 HVGs before declining at larger sizes.

7 Notes

- All runs use the same architecture: latent dim 128, generator layers 256–512–1024, critic layers 1024–512–256, labeler layers 2000–2000–2000, batch size 1024, 20 steps per phase.
- The causal GAN trains in two phases: first the causal controller (a small GAN), then the full causal GAN (generator, critic, labeler, antilabeler). The controller uses minimal GPU memory; the generator phase dominates memory usage.
- Included runs completed within the 80 GB GPU budget. The excluded 17,000 HVG run hit CUDA out-of-memory during generator initialization.
- Generator parameter counts scale linearly with HVG count ($R^2 = 1.0000$); generator step times also scale roughly linearly ($R^2 = 0.9984$, see Figure 3).
- The fitted generator requirements are $\widehat{M}_{\text{gen}}(x) = 4.7268x + 927.7$ MiB and $\widehat{t}_{\text{gen}}(x) = 0.06897x + 15.49$ ms/step ($R^2 = 0.9998$ and $R^2 = 0.9984$, respectively).
- GPU memory values are in MiB as reported by `nvidia-smi`. The GPU total capacity is 81,920 MiB (80 GB).